



Humana-Mays Healthcare Analytics 2019 Case Competition

PREDICTIVE MODEL FOR LONG TERM OPIOID THERAPY

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Introduction

According to recent studies, the number of individuals requiring medical intervention resulting from the overuse of prescription opioids has increased by as much as 153%. This is not surprising given that half of all hospitalized patients receive opioids, a highly addictive drug, for acute pain relief. This problem is further compounded due to many physicians reporting that they believe their opioid prescription training is inadequate and that they lack crucial information regarding which factors contribute to a patient's susceptibility to long-term opioid therapy.

The goal of the following report is to help shed some light on this matter by attempting to identify Humana insurance members that are at risk of long-term opioid use so that early intervention alternatives can be implemented. In doing so, this report aims to prevent members from suffering the consequences of long-term opioid use (both physical anguish and monetary expenses) as well as save Humana the expense of member treatment resulting from complications/conditions due to the overuse of prescription opioids. To accomplish this task, a statistical analysis was conducted in order to predict, beyond baseline accuracy, if a member would continue opioid therapy six months after initial prescribing.

KEY PERFORMANCE INDICATOR

At the onset of the analysis, it was important to first understand the definition of a successful model. Our task was to determine whether a member would be receiving long-term opioid therapy after initial prescribing. LTOT, as defined in the problem scope, was the term used to identify an event in which a member had opioids on hand for at least 90% of the following 180 days after an Opioid Naïve event. An Opioid Naïve event was then defined as an initial opioid prescription when, according to records, the member did not have in their possession an opioid in the preceding 90-day period. A successful model then, would be one that could reliably predict a LTOT event while simultaneously reliably predicting a non-LTOT event, following an Opioid Naïve event. It was important to make this distinction since one of the assumptions we began the analysis with was that they were equally important events of interest. With many binary classification models, we have the ability to produce an increased accuracy of one outcome at the expense of the other if we consider the tradeoff worthwhile; and in this particular situation, we needed to consider the alternatives. On one hand, we could have favored the LTOT event by minimizing its misclassification and producing a model that was exceptional at predicting this event, however this model would wrongly misclassify non-LTOT events as LTOT events to a much greater extent. In practice, this model would likely accomplish the goal of finding members at risk of LTOT and allow for preventative measures to be taken, but at the expense of forcing too many members that are not at risk of LTOT to undergo such measures and potentially miss out on the benefits of opioid use for short-term, acute pain management. On the other hand, we could have favored the non-LTOT event by minimizing its

misclassification, but this would likely result in a model that failed to correctly predict a vast majority of LTOT events, thus defeating the purpose of the analysis all together. Thus, a successful model was defined to be one that could accurately predict both events above and beyond the naïve prediction of classifying each event as belonging to the majority class (non-LTOT). For this reason, a model was deemed successful based on its ROC/AUC score, a measure of its ability to distinguish between the events, and it was the maximization of this metric that guided our analysis.

DATA PREPARATION

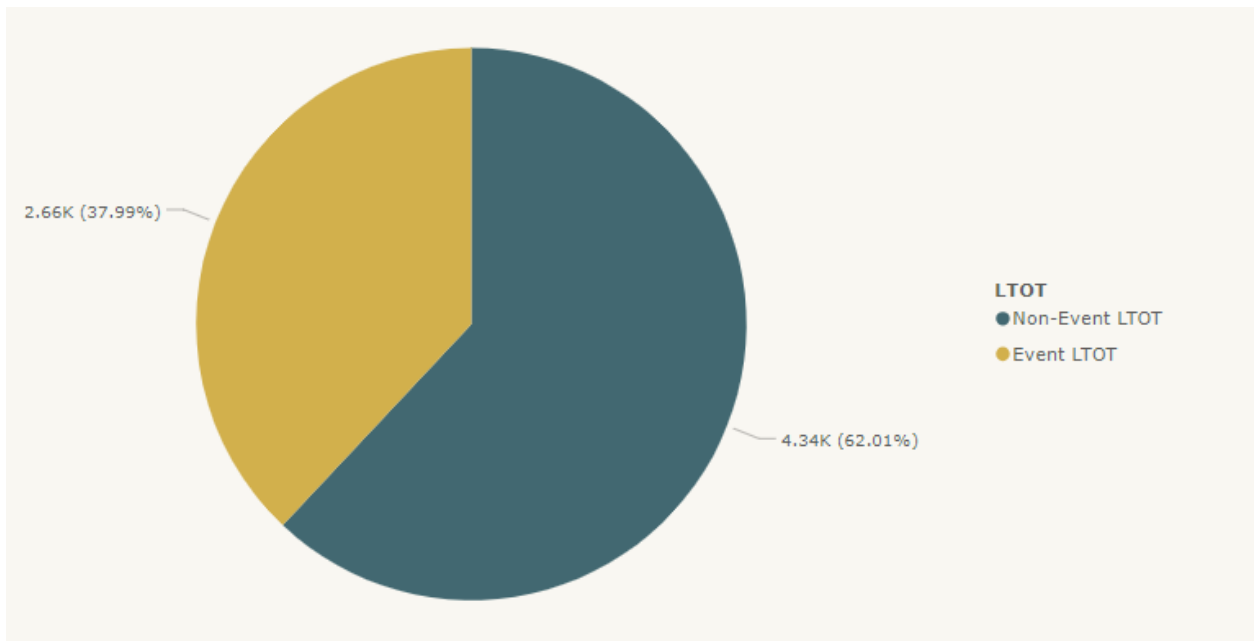
Initial inspection of the data set to be used in modeling, as well as the holdout set to be scored, revealed what appeared to be the combination of several tables from a relational database. We came to this conclusion based on the fact that multiple columns, namely 'event_attr1' through 'event_attr10', held different data types depending on the value in the 'event_descr' column. This posed a considerable challenge to us since we initially did not understand how to get the data into a form that could be used for analysis. Also, many columns which appeared to be missing data were instead simply superfluous, empty cells that resulted from the joining of several database tables. We decided that we needed to somehow transform the data set in such a way as to represent all the data related to a unique ID in a single record. Our best answer for dealing with this dilemma was to add columns that represented the unique values at each intersection of 'event description' and 'event attribute', and then sum or count (depending on data type) the number of instances of each intersection per unique ID. The result would then be a data set containing a single row for each unique ID (14,000 in the model set) and several hundred, perhaps, even thousands of columns representing independent variables to be analyzed. This approach would also have the added benefit of reducing the need for imputation of missing values since, if a value was indeed missing, it was simply not added to the aggregated value. It should be noted that, in addition to these variables, we included a ratio column for each interval variable where the sum for a unique ID was divided by the total days on file in order to fairly compare members with many days of information to members with few days of information.

Given that our team lacked specific domain knowledge in this area, it was our intention from the beginning to include all potential combinations in the analysis so that the insignificant variables could be statistically eliminated from the final model. However, upon execution of this approach, we encountered several roadblocks, mainly computational constraints, that prevented us from completely realizing the data set we intended to produce. In the end, we were forced to not only omit many event description/attribute combinations, but also half of the unique IDs available to us for model formulation. Furthermore, again due to computational constraints, we were limited to records containing an event description of 'RX Claim – Paid'.

DATA ANALYSIS

Data Exploration

After Data Manipulation, we selected a sample of 7000 rows with 210 variables. The target is a binary variable “**LTOT**” where “1” is LTOT event and “0” is Non-LTOT event to opioid.



We don't have missing values in our dataset because we handled missing values in data manipulation.

We partitioned the data into (80/20)% split. For training the data we used 80% and for testing the data we used the remaining 20%.

Data Modeling

First thing we did was select stepwise regression and looked at the variables it selected. And we selected only 13 variables out of 210 variables. After selecting those 13 variables we used logistic regression and stepwise regression. The best auc we could get was 0.85. The auc index seemed pretty good for prediction purposes. There were many outliers in the dataset and we removed them with values greater than (+/-) 3 standard deviation. The data was skewed, we transformed the data with “Max Normal”, “Standardization”, “log 10” method. Treating outliers and transforming variables did not seem to improve the auc index. Sometimes, it even decreased the value of auc index. Here we noticed removing outliers and transforming variables did not improve our auc index.

We tried different methods for variable reduction, one method was LASSO (Least Absolute Shrinkage and Selection Operator). This will allow Betas coefficients to go to zero. LASSO is not only a shrinkage technique, but also a variable selection technique. With this technique we were able to bump up our auc index to 0.87. This result was promising so we tried ELASTIC NET, the concept is the same as the LASSO except, now we are combining both LASSO and Ridge Regression. The result was very poor, the auc index dropped to 0.52.

Second, we now know that trying to fix the outliers and transforming variables did not improve the auc index for the dataset we used. Now, we decided to use decision trees because they are more robust to outliers, missing values and we don't need to normalize the data. Decision trees are easy to interpret as well. With decision tree with 'Entropy' index the auc index was 0.872, it was the same as LASSO. And we tried decision tree with 'Gini' index the auc index was 0.874. This was a slight improvement and the best auc index we had so far.

Third, so far we were getting positive results with the decision tree for our analysis. We used "gini" and "entropy" index and the results were impressive. We wanted to increase the complexity of the decision trees and decided to run random forest for the analysis. The result was very impressive which gave auc index of 0.89. **We finalized 'Random Forest' to be used for modelling the data.**

The parameters we changed in Random Forest are as follows:

n_estimators = 144, random_state = 12345, max_features = 0.7, max_depth = 31, max_leaf_nodes = 33, min_samples_split = 5, min_samples_leaf = 1, n_jobs=-1 and verbose = 1.

The confusion matrix:

	Predicted		
Actual		1	0
	1	(TP) 405	(FN) 130
	0	(FP) 127	(TN) 738

From confusion matrix:

Accuracy = 81.64%, Sensitivity = 75.70%, and Specificity = 85.32%.

With random forest we scored the holding set. As per the instruction we needed the probabilities in order, but the problem was that most of the probabilities of individual IDs were the same so we could not rank them properly. After some discussion, it was expected to have the same probabilities for different IDs because if they fall under the same node, they are expected to have the same probability of the node.

Then we decided to run random forest model 10 times, optimize the parameters and then score the holdout set. Every 10 iteration of random forest we used different random seed. And then we averaged the 10 probabilities with different random seed, which reduced the random seed bias. Even the average probability value had the same probabilities so we could not rank them properly. Then we ran a logistic regression model and scored the holdout set to get the probabilities of individual IDs. Then we did a weighted average between the random forest and logistic regression. The weights assigned to random forest was 0.995 and logistic regression was 0.005. This gave different probabilities for each individual IDs and some strength of the logistic regression as well.

Like we discussed earlier we had 210 variables, and we created a table that shows the important variables that determining LTOT. Top 10 variables that contributed the most are shown below:

Features	Importance
PAY_DAY_SUPPLY_CNT	0.545025
PDSC_RATIO	0.161235
PAYABLE_QTY	0.105605
EVENT_ATTR3	0.040817
PQ_RATIO	0.023572
EVENT_ATTR3_RATIO	0.012486
IR	0.011997
EVENT_ATTR4	0.010658
EVENT_ATTR9	0.008848
Anesthesiology Pain Medicine	0.008715

CONCLUSION

One of the hypotheses for any predictive model is that it is useful only if the model outperforms the baseline model. As per the naïve rule if we categorize all the individual as majority class (i.e. $LTOT = 0$; $LTOT = 0$ is the majority class in the test set and represents 62% of the records) then our model accuracy would be 62%. The random forest performed better than the naïve model with an **accuracy level** of 81.643% (AUC: 0.89, **False Alarm Rate:** 24.3%).

RECOMMENDATIONS

The idea behind this model is to aid medical practitioners, physicians in evaluating if prescribing opioids can cause opioid addiction based on patient's medical history. So, our primary target consumer of this model would be insurance companies, doctors, physicians, surgeons, medical practitioners. This model will be developed into is a brief, self-report screening tool designed to assess risk for opioid abuse among individuals prescribed opioids for various treatments. Patients categorized as high-risk with probability >0.50 are at increased likelihood of future abusive drug-related behavior. Model input will contain information about patient's name, insurance policy number that can be tied back to the patient id from client's database shared by the insurance companies and the output will be probability of becoming a long-term opioid user. With the help of this model, doctors or practitioner can then take a call to "approve", "disapprove" or "approved with other considerations" the opioid purchase in real time to make the decision. A great concern in the study of opioid is that doctors and physicians not always aware how much opioid to prescribe because the chances of opioid addiction may vary from person to person.

Finally, it is a win-win situation for all parties involved – doctors, patients, and the insurance companies like Humana. Doctors, physicians, and hospitals will have a data-based decision model at their fingertips. In turn, Humana would have to pay less by substantially reducing the costs of dealing with insurance claims made by opioid users due to prolonged substance abuse. Additionally, certain patients with $LTOT=1$ probability >0.40 or ≤ 0.50 may be charged higher for their opioid prescription. From the patient's perspective this will discourage them to continue to purchase these drugs after treatment is over. Similarly, this segmentation potentially opens a new revenue channel for Humana. A simple cost-benefit analysis based on classification matrix indicates a **savings of \$30,600 for 1400 patients** with hypothetical costs associated with each scenario.

Cost Benefit Analysis - Confusion Matrix

Cost Matrix	Predicted Class		
	c(i j)	1	0
Actual Class	1	-300	300
	0	60	60

Random Forest	Predicted Class		
	c(i j)	1	0
Actual Class	1	405	130
	0	127	738

$$\text{Accuracy} = (405 + 738) / (405 + 738 + 130 + 127) = 81.643\%$$

$$\text{Cost} = 405 * (-300) + 130 * (300) + 127 * (60) + 738 * (60)$$

$$\text{Cost} = -30,600$$

Costs explanation

Avg. cost of LTOT insurance claim = \$300

Avg. savings by predicting from LTOT insurance claim = -\$300

Avg. cost of LTOT prescription claim = \$60

Avg. cost of patient claim prescription from other network = \$60

LIMITATIONS

However, like any predictive models this model has certain limitations. The available dataset contains only patients registered with Humana. In case, the patient has multiple insurance coverages or buys opioid from black market, there is no way to track that consumption. Secondly, this model assumes that all the patients included in the model have no opioid on hand 90 days prior to the date of prediction. Finally, the success of this model largely depends on current legislations on data sharing and privacy policies. This model works only if we can reserve confidentiality of our patients and yet at the same time isolate patients with the greater health risk from the addiction to opioid.

Further research should be conducted in order to potentially improve model accuracy. Since our computational efficiencies were limited, it stands to reason that increasing the number of records used in the model construction could increase ROC/AUC beyond the capability of this analysis. In addition, sensitive information pertaining to the identity of the individuals used in the analysis, while withheld for the purposes of this study, could be introduced as potentially significant predictor variables capable of increasing model ROC/AUC.

Finally, given the analysis was carried out on an oversampled data set, and the true prevalence of LTOT occurrence in the population is only 1.04%, the accuracy of this model should be further tested and evaluated using a sample containing the true proportion before it can be considered adequate for use in practice.